CSE252 - Introduction to Artificial Intelligence and Machine Learning | Section K3P25RB | CA-1 Groups and Projects

Find your group by your name or registration number. The objectives are what your project will be assessed on; the advanced extension is optional. Use the official source to read the original problem statement (for Smart India Hackathon problems, search the PS ID on the linked page).

Group	Students	Project Title	Problem to Solve	Who Benefits	Objectives (what you will be assessed on)	Suggested Techniques	Syllabus Topics Used	Data to Use	Optional Advanced Extension	Inspired By (official source)
G1	Ayushman (12516219) Vivek Kumar (12526963) Piyush Kumar (12506037)	**Rural Diabetic Retinopathy Screening Assistant with Referral Explanations**	India has the second largest diabetic population and roughly one ophthalmologist per hundred thousand rural people, so retinopathy is caught after vision is already lost.	District hospitals, rural screening camps, ophthalmology referral services	1. Train a retinopathy grader on the APTOS fundus dataset with a patient-level split. 2. Report per-grade sensitivity and quadratic weighted kappa against a logistic-regression baseline. 3. Select and justify a referral threshold at a stated sensitivity target. Produce Grad-CAM overlays for 20 referred cases and ship a Streamlit screening app.	CNN image classification; transfer learning; class imbalance handling; threshold selection from the ROC curve	Unit III-IV neural networks and CNN; Unit V model evaluation, ROC and threshold choice; Unit II image preprocessing and augmentation	APTOS 2019 retinal fundus images (Kaggle) (Find and cite suitable public data yourselves.)	Quantify accuracy loss on deliberately degraded (blurred, under-exposed) images to simulate portable field cameras.	SMART INDIA HACKATHON 2026 PS ID: SIH26038 Explainable AI for Diabetic Retinopathy Screening in Rural India Organisation: MathWorks https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G2	Satyam Shandiliya (12515994) Kale Basweshwar Shantiling (12529711)	**Chest Radiograph Triage System for Emergency Department Queues**	Chest X-rays queue in order of arrival, so a film showing pneumonia waits behind routine studies and treatment is delayed.	Emergency departments, radiology reporting units	1. Train a pneumonia detector on RSNA chest radiographs producing boxes with confidence. 2. Report precision, recall and mAP at IoU 0.5 against a whole-image classifier baseline. 3. Generate a reordered reporting worklist for a held-out set and measure urgency ranking quality. Deploy the detector as a FastAPI service returning boxes plus the ranked worklist.	CNN classification then bounding-box detection; transfer learning; non-maximum suppression	Unit III-IV CNN; Unit V detection metrics and validation; Unit II image preprocessing	RSNA chest radiographs with bounding-box annotations	Compare a two-stage detector against a single-stage one and report the speed/accuracy trade-off.	KAGGLE RSNA Pneumonia Detection Challenge Organizer: Radiological Society of North America (RSNA), hosted on Kaggle https://www.kaggle.com/competitions/rsna-pneumonia-detection-challenge Dataset access: Competition dataset (Kaggle account + rules acceptance)
G3	Preejith Prakashan (12522861) Vishal Krishna M (12501233)	**Screening Mammography Recall Support System**	Screening mammography programmes recall a large number of healthy women for further tests because readers cannot be certain from the image alone.	Breast screening programmes, radiologists, recalled patients	1. Train a malignancy model on RSNA screening mammograms with patient-level aggregation. 2. Report PR-AUC and sensitivity at fixed specificity against a no-imbalance-handling baseline. 3. Calibrate the output probabilities and show the calibration curve before and after. Quantify the recall rate implied by the chosen operating point and dashboard it.	CNN classification on medical images; severe class imbalance handling; probability calibration	Unit III-IV CNN; Unit V imbalanced evaluation, PR curves, calibration; Unit II preprocessing	RSNA screening mammography DICOM studies	Patient-level aggregation across multiple views rather than scoring single images.	KAGGLE RSNA Screening Mammography Breast Cancer Detection Organizer: Radiological Society of North America (RSNA), hosted on Kaggle https://www.kaggle.com/competitions/rsna-breast-cancer-detection Dataset access: Competition dataset (large DICOM; Kaggle account)
G4	Jinadev R P (12515059) Rayapati Mardhav (12513484)	**ICU EEG Monitoring Assistant for Harmful Brain Activity**	Critically ill patients are monitored with continuous EEG that only a trained neurophysiologist can read, so harmful patterns go unnoticed overnight.	Intensive care units, neurology teams	1. Build an EEG classifier over the HMS harmful-activity categories from spectrogram features. 2. Report per-class recall and macro-F1 against a band-power plus random-forest baseline. 3. Define an alerting rule and state its false-alarm rate per monitored hour. Deliver a scrolling EEG monitor with a colour-coded alert bar.	Time-series feature extraction; spectrogram plus CNN; multi-class classification; handling annotator disagreement	Unit III-IV neural networks and CNN; Unit V multi-class metrics; Unit II signal preprocessing and feature engineering	EEG recordings with expert consensus annotations	Model expert disagreement explicitly by predicting the annotator vote distribution rather than a single label.	KAGGLE HMS - Harmful Brain Activity Classification Organizer: Harvard Medical School (HMS), hosted on Kaggle https://www.kaggle.com/competitions/hms-harmful-brain-activity-classification Dataset access: Competition dataset (Kaggle account)
G5	Kanishak Patyal (12523884) Ruhani Sharma (12514000)	**Twelve-Lead ECG Abnormality Screening for Primary Health Centres**	Primary health centres record ECGs but have no cardiologist to read them, so abnormalities are found only after referral.	Primary health centres, rural cardiology referral pathways	1. Train a multi-label 12-lead ECG classifier on the PhysioNet 2020 Challenge data. 2. Report per-class F1 and macro-F1 against a rule-based interval-threshold baseline. 3. Define a referral rule for urgent abnormalities and measure its recall. Ship a Streamlit app that displays the waveform with predicted labels.	Multi-label classification; 1D CNN on raw signal; signal filtering and segmentation	Unit III-IV neural networks; Unit V multi-label evaluation; Unit II signal preprocessing	12-lead ECG recordings with cardiac abnormality labels	Compare a 1D CNN on raw signal against handcrafted clinical features and report which wins per class.	PHYSIONET Classification of 12-lead ECGs: The PhysioNet/Computing in Cardiology Challenge 2020 Organizer: PhysioNet with Computing in Cardiology https://physionet.org/content/challenge-2020/1.0.2/ Dataset access: Public and directly downloadable
G6	Patnana Mahitesh (12509827) Nagireddy Bhagath Chandra (12521738)	**Early Literacy Screening Assistant for Primary Classrooms**	Reading difficulties are identified far too late because one-to-one literacy screening requires teacher time that schools do not have.	Primary schools, teachers, special-education coordinators	1. Build a literacy-screener scorer from MFCC features on the DrivenData audio clips. 2. Report log loss and ROC-AUC on held-out speakers against a majority-class baseline. 3. Compare handcrafted MFCC features against a pretrained speech embedding. Deliver a teacher-facing tool that scores a clip and flags children for review.	Audio feature extraction (MFCC); binary classification; handling noisy child speech	Unit III-IV neural networks; Unit V classification metrics and calibration; Unit II audio feature engineering	Audio clips from early literacy screeners	Compare handcrafted MFCC features against a pretrained speech embedding.	DRIVENDATA Goodnight Moon, Hello Early Literacy Screening Organizer: DrivenData, with literacy screening partners https://www.drivendata.org/competitions/298/literacy-screening/ Dataset access: Competition dataset (DrivenData account)
G7	Rishav Raj (12523073) Inpreet Kaur (12519786)	**Retail Replenishment Forecasting and Stock-Out Cost Dashboard**	Retailers commit to stock before demand is known, and a single error figure hides which products and days actually cause lost sales.	Retail chains, category managers, warehouse planners	1. Build a hierarchical demand forecaster on the M5 Walmart sales series. 2. Report MAE, RMSE and MASE against mandatory seasonal-naive and moving-average baselines. 3. Translate forecast error into stock-out and holding cost per product category. Deliver a planning dashboard showing forecast versus actual with the cost impact panel.	Time-series forecasting; lag and calendar feature engineering; gradient boosting; hierarchical aggregation	Unit VI time-series analysis; Unit II feature engineering; Unit V forecast evaluation	Walmart hierarchical unit-sales series	Reconcile forecasts across the product hierarchy so item forecasts sum to department totals.	KAGGLE M5 Forecasting - Accuracy Organizer: M Open Forecasting Center (University of Nicosia), hosted on Kaggle https://www.kaggle.com/competitions/m5-forecasting-accuracy Dataset access: Competition dataset (Kaggle account + rules acceptance)
G8	Aakanksha Rathore (12515077) Ridhima Dhingra (12515671)	**Dengue Outbreak Early Warning System for City Health Departments**	Dengue response begins after case numbers rise, when vector control is already too late to prevent the peak.	Municipal health departments, vector control teams	1. Build a weekly dengue case forecaster for San Juan and Iquitos from climate features. 2. Report MAE against seasonal-average and persistence baselines for both cities. 3. Define a graded alert rule and report its precision and recall on surge weeks. Deliver a dashboard showing the forecast band and the alert timeline.	Time-series regression; lagged climate features; regularised linear and tree models	Unit VI time-series; Unit II feature engineering and lag construction; Unit V regression evaluation	Environmental and climate series for San Juan and Iquitos with weekly dengue case counts	Add a negative-binomial or count-appropriate model and compare against the Gaussian assumption.	DRIVENDATA DengAI: Predicting Disease Spread Organizer: DrivenData, using data from U.S. Federal Government agencies https://www.drivendata.org/competitions/44/dengai-predicting-disease-spread/ Dataset access: Public practice competition (DrivenData account, freely downloadable)
G9	Jiya Soni (12515185) Nimrat Kaur (12502147)	**Vaccination Uptake Targeting System for Immunisation Campaigns**	Immunisation campaigns spend outreach budget uniformly although uptake varies sharply by attitude, access and demographics.	Public health agencies, immunisation programme managers	1. Train calibrated H1N1 and seasonal-flu uptake models on the DrivenData survey data. 2. Report mean ROC-AUC across both targets against a logistic-regression baseline. 3. Produce a ranked outreach list with the drivers behind each ranking. Audit whether the model reconstructs protected characteristics as proxies and report it.	Multi-target binary classification; categorical encoding; feature importance analysis	Unit III supervised classification; Unit II encoding and missing values; Unit V ROC-AUC and interpretation	National 2009 H1N1 Flu Survey respondent records	Add SHAP explanations and test whether the model reproduces protected characteristics as proxies.	DRIVENDATA Flu Shot Learning: Predict H1N1 and Seasonal Flu Vaccines Organizer: DrivenData, using U.S. National 2009 H1N1 Flu Survey data https://www.drivendata.org/competitions/66/flu-shot-learning/ Dataset access: Public practice competition (DrivenData account, freely downloadable)
G10	Kunal Sandha (12521234) Ishaan Thakur (12526736)	**Village-Level Livestock Disease Early Detection and Alert Platform**	Livestock disease is reported late and diagnostic facilities are distant, so outbreaks spread through a district before any response begins.	Para-veterinary workers, district animal husbandry departments, livestock owners	1. Build a symptom-to-risk classifier for livestock disease from public outbreak records. 2. Add a spatio-temporal alert rule over village-week report clusters. 3. Report alert precision, recall and days of warning against a fixed case-count threshold. Deliver a village-level map dashboard with the alert overlay.	Symptom classification; spatio-temporal anomaly detection; clustering of report patterns	Unit III classification; Unit VI clustering and anomaly detection; Unit V precision/recall for rare events	FAO EMPRES-i animal disease outbreak records (public) (Find and cite suitable public data yourselves.)	Add a mobile-friendly offline-first reporting form and multilingual symptom entry.	SMART INDIA HACKATHON 2026 PS ID: SIH26128 Efficient systems for early detection, prevention, and management of livestock diseases and animal health issues Organisation: Government Of Maharashtra https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G11	Maru Snehith Reddy (12505462) Shourya Mittal (12505875)	**Field-Deployable Crop Disease Advisory for Smallholder Farmers**	Smallholder farmers recognise disease only after visible damage spreads, and laboratory diagnosis is neither near nor timely, which leads to wrong or excessive pesticide use.	Smallholder farmers, agricultural extension officers	1. Train a crop disease classifier with an abstention rule for low-confidence images. 2. Report macro-F1 and the coverage-accuracy curve against a colour-histogram plus SVM baseline. 3. Measure the accuracy drop from laboratory images to real field photographs and state it. Deliver a mobile-friendly advisory app returning diagnosis, confidence and recommended action.	CNN image classification; transfer learning; abstention on low confidence; data augmentation	Unit III-IV CNN and transfer learning; Unit V confidence and abstention; Unit II augmentation	Cassava Leaf Disease Classification images (Kaggle), plus PlantDoc field imagery (public) (Find and cite suitable public data yourselves.)	Measure the accuracy drop from clean laboratory images to real field photographs and report it honestly.	SMART INDIA HACKATHON 2026 PS ID: SIH26131 Early detection and management of crop diseases and pest infestations Organisation: Government Of Maharashtra https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data

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G12	Pavan Pampana (12519780) Janke Venkata Gopalakrishna Sasidhar Reddy (12518565)	**Cassava Disease Identification Under Real Farm Photography Conditions**	Leaf disease models report near-perfect laboratory accuracy and then fail on the blurred, shadowed, cluttered photographs farmers actually take.	Smallholder cassava farmers, agricultural NGOs	1. Train a cassava disease classifier on the Kaggle competition images. 2. Build an image-quality gate that refuses unusable photographs and prompts a retake. 3. Report accuracy on gated versus ungated subsets against a single-CNN no-gate baseline. Deliver a camera-input app that demonstrates the retake prompt working.	CNN classification; transfer learning; image quality assessment; augmentation	Unit III-IV CNN; Unit II image preprocessing and augmentation; Unit V evaluation under distribution shift	Cassava leaf image dataset	Test-time augmentation and an ensemble, with the added cost reported against the accuracy gained.	KAGGLE Cassava Leaf Disease Classification Organizer: Hosted on Kaggle (sponsoring institution stated on the official page - VR) https://www.kaggle.com/competitions/cassava-leaf-disease-classification Dataset access: Competition dataset (Kaggle account + rules acceptance)
G13	Yash Rai (12521744) Anmol Kainth (12501717)	**Wheat Head Counting System for Yield Estimation Across Growing Regions**	Yield estimation depends on manual head counts in sample plots, which is slow, subjective and impossible to scale across regions.	Agricultural research stations, crop insurers, state agriculture departments	1. Train a wheat head detector on the Global Wheat Detection dataset. 2. Report mAP at IoU 0.5-0.75 and plot-level count MAE against a classical blob-detection baseline. 3. Evaluate cross-region generalisation by holding out entire growing regions. Deliver a tool returning an annotated plot image with a count and uncertainty range.	Object detection; bounding-box regression; non-maximum suppression; density estimation	Unit III-IV CNN and detection; Unit V detection metrics; Unit II augmentation	Global Wheat Head Detection images with bounding boxes	Test cross-region generalisation by holding out entire growing regions from training.	KAGGLE Global Wheat Detection Organizer: Global Wheat Head Detection collaboration, hosted on Kaggle https://www.kaggle.com/competitions/global-wheat-detection Dataset access: Competition dataset (Kaggle account + rules acceptance)
G14	Chandramouli Kandregula (12513030)	**Non-Destructive Root Volume Estimation for Crop Breeding Trials**	Root mass is the trait breeders most want and least often measure, because measuring it means digging the plant up and destroying the trial.	Crop breeding programmes, agricultural research institutes	1. Train a CNN regressor predicting root volume from CGIAR underground scan images. 2. Report RMSE, MAE and R-squared against an image-area heuristic regression baseline. 3. Produce prediction intervals and verify their empirical coverage. Deliver a tool returning a volume estimate with a confidence band.	CNN regression on images; data augmentation; regression uncertainty	Unit III-IV CNN; Unit V regression metrics; Unit II image preprocessing	Underground root scanning images with measured root volume	Predict an interval rather than a point estimate and verify its empirical coverage.	ZINDI CGIAR Root Volume Estimation Challenge Organizer: CGIAR, hosted on Zindi https://zindi.africa/competitions/cgiar-root-volume-estimation-challenge Dataset access: Competition dataset (Zindi account)
G15	Shine Kaur (12520300) Akshadha Awasthi (12512354)	**Forest Biomass Mapping System for Carbon Accounting**	Carbon accounting needs biomass estimates over whole forests, but the LiDAR surveys that measure it accurately cover only a fraction of the area.	Forest departments, carbon registries, conservation agencies	1. Train a biomass regressor on Sentinel imagery against the BioMasters LiDAR targets. 2. Report RMSE (the official metric) against a vegetation-index linear-regression baseline. 3. Break error down by forest density band and quantify what radar adds over optical. Render an interactive biomass map with per-pixel uncertainty.	Image regression; multi-band satellite feature engineering; CNN or gradient boosting on patches	Unit III-IV neural networks; Unit II multi-band feature engineering; Unit V regression evaluation	Sentinel-1 and Sentinel-2 imagery with LiDAR-derived biomass targets for Finnish forests	Combine radar and optical bands and quantify what each modality contributes.	DRIVENDATA The BioMasters Organizer: DrivenData, with the Finnish Forest Centre https://www.drivendata.org/competitions/99/biomass-estimation/ Dataset access: Competition dataset (DrivenData account)
G16	Rishabh Pradeep Singh (12524249) Kunwarpal Singh (12527518)	**Next-Day Air Quality Forecasting and Advisory System for Delhi NCR**	Air quality advisories are issued once pollution has arrived, when a day of warning would let schools, hospitals and vulnerable people act.	State pollution control boards, schools, hospitals, city administrations	1. Build a next-day AQI forecaster for Delhi NCR from coupled weather and pollutant data. 2. Report MAE and RMSE against a mandatory persistence baseline. 3. Report advisory-band precision and recall separately for severe-pollution days. Deliver a dashboard with the forecast, advisory band and station map.	Time-series forecasting; meteorological feature engineering; regression and classification of AQI bands	Unit VI time-series; Unit II feature engineering; Unit V forecast evaluation against baselines	CPCB station data and OpenAQ (public, API) (Find and cite suitable public data yourselves.)	Model the weather-pollution feedback explicitly (inversion, boundary-layer proxies) and report the gain.	SMART INDIA HACKATHON 2026 PS ID: SIH26082 Air Pollution-Weather Coupled Forecasting System (Delhi NCR Focus) Organisation: Ministry of Earth Sciences (MoES) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G17	Godwin Poulose (12518036) Muhammed Shazin P C (12526212)	**Aviation Contrail Detection System for Climate Impact Reduction**	Aircraft contrails contribute materially to warming and could be avoided by small routing changes, but nobody can currently detect them at scale.	Airlines, air traffic management, climate research groups	1. Train a contrail segmentation model on the Google Research satellite imagery. 2. Report Dice and IoU against a brightness-temperature thresholding baseline. 3. Quantify the gain from using the temporal image sequence over a single frame. Deliver a viewer overlaying the predicted contrail mask on the satellite scene.	Semantic segmentation; U-Net style CNN; handling thin elongated structures and heavy class imbalance	Unit III-IV CNN; Unit V segmentation metrics; Unit II multi-band image preprocessing	GOES-16 satellite imagery with contrail masks	Use the temporal image sequence rather than a single frame and report the improvement.	KAGGLE Google Research - Identify Contrails to Reduce Global Warming Organizer: Google Research, hosted on Kaggle https://www.kaggle.com/competitions/google-research-identify-contrails-reduce-global-warming Dataset access: Competition dataset (Kaggle account + rules acceptance)
G18	Abhiram Chandran (12514511) Ashin S Krishna (12512103)	**Harmful Algal Bloom Warning System for Inland Water Bodies**	Cyanobacteria blooms poison drinking water and kill livestock and pets, and are usually confirmed only after someone falls ill.	Water utilities, public health departments, lake management authorities	1. Build a cyanobacteria severity estimator from satellite imagery on the Tick Tick Bloom data. 2. Report RMSE (the official metric) against mean-predictor and band-index heuristic baselines. 3. Hold out entire geographic regions and report the spatial generalisation gap. Deliver a map dashboard with severity pins and the public advisory text.	Image regression and severity classification; satellite band feature engineering; spatial train/test splitting	Unit II feature engineering; Unit III classification and regression; Unit V evaluation with spatial holdout	Satellite imagery with in-situ cyanobacteria measurements	Hold out entire geographic regions to test spatial generalisation honestly.	DRIVENDATA Tick Tick Bloom: Harmful Algal Bloom Detection Challenge Organizer: DrivenData (NASA-supported challenge) https://www.drivendata.org/competitions/143/tick-tick-bloom/ Dataset access: Competition dataset (DrivenData account)
G19	Bhavya Yadav (12518541) Sanchi Sharma (12505826)	**Rural Drinking Water Safety Assessment from Incomplete Test Records**	Water testing in rural and mining-affected districts is patchy, with several measurements missing from most samples, yet assessment tools assume complete records.	Gram panchayats, district water departments, community health workers	1. Build a potability classifier that operates on incomplete water test records. 2. Compare at least three missing-data strategies against a complete-case-analysis baseline. 3. Report unsafe-class recall and the performance-versus-missingness curve. Add a next-test recommendation naming the measurement that most reduces uncertainty.	Classification with missing data; imputation strategy comparison; feature importance for test prioritisation	Unit II missing-value handling; Unit III classification; Unit V evaluation under degradation	UCI Water Potability and CPCB water quality data (public) (Find and cite suitable public data yourselves.)	Add a cost-aware recommendation that balances test cost against uncertainty reduction.	SMART INDIA HACKATHON 2026 PS ID: SIH26040 Smart Water Purification and Quality Monitoring System for Rural and Mining-Affected Areas Organisation: Government of Jharkhand https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G20	Vaishnavi Singh (12518715) Isha Rani (12517595)	**Rural Water Point Failure Prediction for Maintenance Scheduling**	Thousands of rural water points are broken at any time and repair crews find out only when a village reports it, often months later.	Rural water agencies, maintenance crews, village committees	1. Train a water point functional-status classifier on the Tanzanian waterpoint records. 2. Report classification rate and per-class F1 against majority-class and single-tree baselines. 3. Produce a ranked maintenance list and report precision at the top K visits. Deliver a map with colour-coded water points and a crew route list.	Multi-class classification; categorical encoding of high-cardinality fields; tree ensembles	Unit II encoding and missing values; Unit III decision trees and random forests; Unit V multi-class metrics	Tanzanian waterpoint records	Add geospatial clustering so crews receive routed batches rather than a flat list.	DRIVENDATA Pump it Up: Data Mining the Water Table Organizer: DrivenData, with Taarifa and the Tanzanian Ministry of Water https://www.drivendata.org/competitions/7/pump-it-up-data-mining-the-water-table/ Dataset access: Public practice competition (DrivenData account, freely downloadable)
G21	Harmandeep Kaur (12508749) Madhav Bhanot (12510855)	**Flood Extent Mapping from Radar Imagery for Emergency Response**	After a flood, responders cannot see through cloud cover with optical satellites, so they do not know which areas are actually underwater when it matters most.	Disaster management authorities, relief agencies, district administrations	1. Train a floodwater segmentation model on Sentinel-1 SAR imagery. 2. Report IoU and Dice against the classical SAR backscatter thresholding baseline. 3. Overlay population density to convert flood extent into an exposure estimate. Deliver an interactive map showing the flood mask and affected settlements.	Semantic segmentation; SAR image preprocessing; handling speckle noise; class imbalance	Unit III-IV CNN; Unit V segmentation metrics; Unit II image preprocessing	Sentinel-1 synthetic-aperture radar (SAR) imagery with flood labels	Overlay population density to convert flood extent into an exposure estimate for prioritisation.	DRIVENDATA STAC Overflow: Map Floodwater from Radar Imagery Organizer: DrivenData, with Microsoft AI for Earth and Cloud to Street https://www.drivendata.org/competitions/81/detect-flood-water/ Dataset access: Competition dataset (DrivenData account)
G22	Palak Gupta (12528440) Nitish Raj (12522052)	**Post-Earthquake Building Damage Triage System**	After a major earthquake, engineers must inspect tens of thousands of buildings one by one to decide which are safe to re-enter.	Disaster response agencies, structural engineers, municipal authorities	1. Train a building damage-grade classifier on the Nepal Gorkha survey records. 2. Report micro-averaged F1 against majority-class and logistic-regression baselines. 3. Compare ordinal against nominal treatment of the damage grade. Deliver a prioritised inspection queue dashboard.	Ordinal multi-class classification; categorical encoding; tree ensembles; feature importance	Unit II encoding and feature engineering; Unit III trees and ensembles; Unit V multi-class evaluation	Nepal building damage survey records	Treat the target as ordinal rather than nominal and show whether it improves the ranking.	DRIVENDATA Richter's Predictor: Modeling Earthquake Damage Organizer: DrivenData, using 2015 Gorkha earthquake survey data https://www.drivendata.org/competitions/57/nepal-earthquake/ Dataset access: Public practice competition (DrivenData account, freely downloadable)
G23	S Pradeep (12509156) Aditya Pandey (12511007)	**Aerial Search and Rescue Assistant for Disaster Zones**	In the first hours after a disaster, responders lack situational awareness and search on foot through terrain that may be impassable.	Search and rescue teams, NDRF, state disaster response forces	1. Train a person and hazard detector on public aerial search-and-rescue imagery. 2. Report mAP at IoU 0.5 and small-object recall against a generic pretrained detector. 3. Implement tiled inference and quantify the recall gain on small objects. Deliver a viewer showing detections and a ranked search-priority map.	Object detection on aerial imagery; small-object detection; detection confidence calibration	Unit III-IV CNN and detection; Unit V detection metrics; Unit II augmentation for scale variation	Public aerial search-and-rescue and person-detection datasets (e.g. SARD, HERIDAL) (Find and cite suitable public data yourselves.)	Add tiled inference for high-resolution frames and report the recall gain on small objects.	SMART INDIA HACKATHON 2026 PS ID: SIH26177 A deployable AI-powered autonomous drone that aids search-and-rescue operations by detecting people and hazards Organisation: Qualcomm Inc https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data

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G24	Mankiran Kaur (12530109) Rohit (12508736)	**Mine Subsidence Early Warning System for Coalfield Communities**	Subsidence above underground coal mines is monitored by periodic manual survey, so ground failure is usually assessed after it has damaged homes and land.	Ministry of Coal, mine operators, coalfield villages	1. Build an unsupervised anomaly detector over ground-deformation monitoring series. 2. Report warning lead time and detection rate against a fixed displacement threshold. 3. State the false alarms per month an operator would have to triage. Deliver a dashboard showing displacement trend with staged warning bands.	Time-series anomaly detection; trend and change-point detection; unsupervised scoring without failure labels	Unit VI time-series and anomaly detection; Unit II preprocessing; Unit V rare-event evaluation	Public InSAR ground-deformation products and subsidence monitoring datasets (Find and cite suitable public data yourselves.)	Add multi-sensor fusion and report which sensor contributes the earliest signal.	SMART INDIA HACKATHON 2026 PS ID: SIH26025 Development of an AI-enabled Low Cost Real Time Mine Subsidence Monitoring, Prediction and Early Warning System for Underground Coal Mines in India Organisation: Ministry of Coal https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G25	Krishna Verma (12517062)	**Multi-Camera Vehicle Trajectory and Traffic Analytics Engine**	Cities have extensive camera networks that process each feed in isolation, so authorities cannot follow a vehicle across the city or extract movement trends.	City traffic police, urban transport authorities	1. Build a plate detection and recognition pipeline and report per-stage accuracy. 2. Report detection mAP and plate-level recognition accuracy against an OCR-only baseline. 3. Link the same vehicle across cameras and measure trajectory linking precision. Deliver a camera-map dashboard showing reconstructed trajectories.	Detection plus OCR; feature matching for re-identification; sequence linking across cameras	Unit III-IV CNN, detection and OCR; Unit V recognition metrics; Unit II image preprocessing	Public ANPR datasets (e.g. CCPD) and Indian plate datasets on Kaggle (Find and cite suitable public data yourselves.)	Add plate-independent vehicle re-identification for cases where the plate is unreadable.	SMART INDIA HACKATHON 2026 PS ID: SIH26127 City-Wide AI Engine for Multi-Camera ANPR Trajectory Tracking and Urban Traffic Analytics Organisation: Bharat Electronics Limited https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G26	Manvendra Singh (12526255) Aarunya Attri (12515061) Gaurav Gulwani (12514423)	**Traffic Anomaly and Incident Detection from City Camera Feeds**	Road incidents are reported by whoever happens to call, so response begins minutes or tens of minutes after the event.	Traffic control rooms, emergency dispatch, highway authorities	1. Build a trajectory-based incident detector over city camera footage. 2. Report precision, recall and time-to-detection against a fixed motion-threshold rule. 3. State false alarms per camera-hour, the number a control room actually cares about. Deliver a control-room style dashboard with a live alert feed.	Object detection and tracking; motion feature extraction; anomaly detection on trajectories	Unit III-IV CNN and detection; Unit VI anomaly detection; Unit V rare-event metrics	AI City Challenge track datasets (registration required)	Add multi-camera consistency checking to suppress single-camera false alarms.	AI CITY CHALLENGE AI City Challenge 2024 - Challenge Tracks Organizer: AI City Challenge (CVPR workshop); track sponsors stated on the official page - VR https://www.aicitychallenge.org/2024-challenge-tracks/ Dataset access: Competition dataset (registration and data agreement required)
G27	Yuvraj Singh (12517178) Mohsin Farooq (12528791)	**Railway Rescheduling Agent for Delay Recovery**	One delayed train cascades across the network, and controllers reschedule manually under time pressure with no way to compare options.	Railway operations control centres	1. Implement a rescheduling agent in the Flatland 3 simulator. 2. Report total delay recovered against mandatory do-nothing and greedy baselines. 3. Measure how recovery degrades as disruption severity increases. Deliver an animated simulation replay with a metrics dashboard.	Multi-agent pathfinding; heuristic search; reinforcement learning basics; simulation	Unit I search and problem solving; Unit VI reinforcement learning basics; Unit V policy evaluation	Flatland railway simulation environment	Compare a learned policy against a classical planner and report where each wins.	AICROWD Flatland 3 Organizer: Alcrowd (railway operator partners stated on the official page - VR) https://www.aicrowd.com/challenges/flatland-3 Dataset access: Open-source simulator (freely installable)
G28	Hari Ramachandran S (12527511) Mirza Ali Baaqar (12516009)	**Urban Road Condition Mapping Using the Public Bus Fleet**	Road condition surveys are expensive and infrequent, while city buses already travel almost every major road every single day.	Municipal corporations, road maintenance departments, transport undertakings	1. Build a road surface defect classifier from vehicle-mounted sensing data. 2. Report per-defect precision and recall against a vertical-acceleration threshold baseline. 3. Aggregate detections to road segments and validate against a manual survey. Deliver a city map with a condition heat-map and a ranked repair list.	Sensor/image feature extraction; classification of surface defects; geospatial aggregation	Unit II feature engineering; Unit III classification; Unit VI clustering for hotspot detection	Public pothole/road damage datasets plus student-collected phone sensor data (Find and cite suitable public data yourselves.)	Fuse camera and inertial signals and report which modality detects which defect type better.	SMART INDIA HACKATHON 2026 PS ID: SIH26124 AI-Powered Mobile Urban Intelligence Platform Using Public Transport Fleet Organisation: Bharat Electronics Limited https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G29	Samarveer Singh Chundawat (12513426)	**Building Energy Consumption Forecasting for Retrofit Verification**	Building owners pay for efficiency retrofits and then cannot prove the savings, because they have no credible estimate of what consumption would have been without the work.	Facility managers, energy service companies, building owners	1. Train per-meter building consumption models on the ASHRAE dataset. 2. Report RMSLE and MAE against a mandatory same-hour-last-week persistence baseline. 3. Break error down by building type and meter type. Deliver a dashboard comparing predicted against actual with a savings panel.	Time-series regression; weather and calendar feature engineering; gradient boosting; per-meter modelling	Unit VI time-series; Unit II feature engineering and missing data; Unit V regression evaluation	ASHRAE building meter readings with weather and metadata	Add uncertainty bands so a savings claim can be stated with confidence rather than as a point figure.	KAGGLE ASHRAE - Great Energy Predictor III Organizer: ASHRAE (American Society of Heating, Refrigerating and Air-Conditioning Engineers), hosted on Kaggle https://www.kaggle.com/competitions/ashrae-energy-prediction Dataset access: Competition dataset (Kaggle account + rules acceptance)
G30	Aditya Raj Singh (12504841) Aashu Kumari (12501861) Shreya Kumari (12509174)	**Prosumer Energy Imbalance Forecasting for Grid Balancing**	Households with solar panels both consume and produce power unpredictably, and the resulting imbalance costs the grid operator real money every day.	Distribution system operators, energy retailers, prosumer households	1. Build separate prosumer production and consumption forecasters on the Enefit data. 2. Report MAE against a previous-day persistence baseline for both targets. 3. Translate forecast error into an imbalance cost in currency terms. Deliver a dashboard tracking forecast and imbalance cost.	Time-series forecasting; weather feature integration; separate production and consumption models	Unit VI time-series; Unit II feature engineering; Unit V forecast evaluation and cost translation	Estonian prosumer production/consumption with weather and prices	Add a client-segment analysis showing which prosumer types are hardest to forecast.	KAGGLE Enefit - Predict Energy Behavior of Prosumers Organizer: Enefit (Eesti Energia), hosted on Kaggle https://www.kaggle.com/competitions/predict-energy-behavior-of-prosumers Dataset access: Competition dataset (Kaggle account + rules acceptance)
G31	Deepshikha Tiwari (12524217) Anmol Kumar (12525001)	**Building Cluster Energy Control Agent for Peak Demand Reduction**	Buildings each optimise their own energy use, and the resulting synchronised demand creates the very peaks that make electricity expensive and dirty.	District energy operators, campus facilities, grid operators	1. Implement a building-cluster control policy in the CityLearn simulator. 2. Report peak demand, total cost and comfort violations against a mandatory rule-based controller. 3. Show the trade-off between peak reduction and comfort explicitly. Deliver load curves before and after control in a simulation dashboard.	Reinforcement learning basics; simulation-based policy evaluation; multi-objective reward design	Unit VI reinforcement learning; Unit I search and optimisation; Unit V policy evaluation	CityLearn building energy simulation environment	Compare a learned agent against a classical rule-based controller and report where each is better.	AICROWD NeurIPS 2022: CityLearn Challenge Organizer: Alcrowd, NeurIPS 2022 competition track https://www.aicrowd.com/challenges/neurips-2022-citylearn-challenge Dataset access: Open-source simulator (freely installable)
G32	Nilesh Gupta (12523427) Ayush Ranjan Nayak (12519503)	**Card Transaction Fraud Screening with Investigator Workload Costing**	Fraud is a fraction of one percent of transactions, so a system that approves everything scores over ninety-nine percent accuracy and catches nothing.	Payment processors, card issuers, fraud investigation teams	1. Train a fraud model on the IEEE-CIS transaction data with engineered entity features. 2. Report ROC-AUC and PR-AUC against logistic-regression and majority-class baselines. 3. Cost both error types and choose the threshold minimising total expected cost. Report the daily investigation queue size that threshold implies and dashboard it.	Imbalanced classification; feature engineering on transaction and identity fields; gradient boosting; threshold optimisation	Unit II feature engineering and encoding; Unit III classification; Unit V imbalanced metrics and threshold selection	IEEE-CIS / Vesta e-commerce transaction data	Add entity-level aggregation features (card, device, email domain) and report the lift they provide.	KAGGLE IEEE-CIS Fraud Detection Organizer: IEEE Computational Intelligence Society (IEEE-CIS) with Vesta, hosted on Kaggle https://www.kaggle.com/competitions/ieee-fraud-detection Dataset access: Competition dataset (Kaggle account + rules acceptance)
G33	Parinoor Kaur (12513477) Rudraksh (12521989)	**Credit Scoring for Thin-File Applicants with Adverse Action Reasons**	Applicants without a credit history are refused by default, and those refused are rarely told what would actually change the decision.	Lenders, credit bureaus, first-time borrowers	1. Train a calibrated repayment model on the Home Credit applicant data. 2. Report ROC-AUC and calibration against a bureau-data-only logistic-regression baseline. 3. Generate plain-language adverse-action reasons for refused applications. Audit error rates across demographic proxies and report what attribute removal does not fix.	Binary classification on wide tabular data; missing-value handling; feature importance and explanation	Unit II missing values and encoding; Unit III classification and ensembles; Unit V ROC-AUC and calibration	Home Credit applicant and bureau records	Audit error rates across demographic proxies and report what removing an attribute does and does not fix.	KAGGLE Home Credit Default Risk Organizer: Home Credit Group, hosted on Kaggle https://www.kaggle.com/competitions/home-credit-default-risk Dataset access: Competition dataset (Kaggle account + rules acceptance)
G34	Daulat Ram (12522431) Mehak Garg (12525175)	**Network Attack Forecasting and Early Warning for Critical Infrastructure**	Signature-based defences recognise only attacks that have already been catalogued, so genuinely new attacks are detected after the damage.	Security operations centres, critical infrastructure operators, NTRO-aligned agencies	1. Build a label-free network anomaly detector on public intrusion datasets. 2. Evaluate on attack classes held out entirely and report detection rate and false positives per hour. 3. Compare against a supervised classifier to quantify what labels are actually worth. Add a short-horizon alert-volume forecast and deliver a SOC-style dashboard.	Unsupervised anomaly detection; traffic feature engineering; time-series forecasting of alert volume	Unit VI anomaly detection and time-series; Unit II feature engineering; Unit V rare-event evaluation	CICIDS2017, UNSW-NB15, NSL-KDD (public, directly downloadable) (Find and cite suitable public data yourselves.)	Compare unsupervised detection against a supervised model and quantify what labels are actually worth.	SMART INDIA HACKATHON 2026 PS ID: SIH26153 AI based Network Attack Forecasting from Network Traffic Data Organisation: National Technical Research Organisation (NTRO) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G35	Lakshya Jain (12513897)	**Multi-Label Toxicity Moderation Assistant with Reviewer Routing**	Moderation teams cannot read everything, and blunt filters silence ordinary speech while missing genuinely harmful content.	Online platforms, community moderators, trust and safety teams	1. Train a multi-label toxicity classifier on the Jigsaw comment data. 2. Report mean column-wise ROC-AUC against a TF-IDF plus logistic-regression baseline. 3. Tune per-label thresholds and define a human review-routing rule with its workload. Measure the false-positive rate on a curated benign set and report the moderation harm.	Multi-label text classification; text vectorisation and embeddings; threshold tuning per label	Unit V NLP fundamentals and text representation; Unit III classification; Unit V multi-label metrics	Wikipedia talk-page comments with toxicity labels	Measure false positives on reclaimed and in-group language and report the moderation harm.	KAGGLE Toxic Comment Classification Challenge Organizer: Conversation AI team (Jigsaw / Google), hosted on Kaggle https://www.kaggle.com/competitions/jigsaw-toxic-comment-classification-challenge Dataset access: Competition dataset (Kaggle account + rules acceptance)